Review Report

Manuscript: *Bayesian Estimation and Forecasting of Internal Migration Rates: An Application to the UK*

Overall assessment and recommendation: Major Revision

This manuscript addresses an important applied problem in demographic estimation and forecasting. The highly disaggregated UK internal-migration data present genuine challenges arising from heterogeneity across age, sex, origin, destination, and time, and the authors construct a flexible Bayesian hierarchical framework that is capable of borrowing information across a very large number of cells. The manuscript is generally clearly written, the progression from the baseline specification to richer interaction structures is easy to follow, and the reported computational feasibility of fitting a large hierarchical model is potentially useful for applied demographic work. The empirical examples also illustrate why flexible age and origin–destination structure can matter in practice.

At the same time, I believe substantial revision is needed before the contribution of the paper can be properly assessed. In particular, the current manuscript does not yet distinguish clearly enough between methodological development and the application/integration of standard Bayesian hierarchical components; the regularization mechanism in the very high-dimensional specification needs a more explicit treatment; the inferential computation requires a clearer description; and the forecasting claims need to be supported by genuine out-of-sample validation. I therefore recommend **major revision**. My main comments are below.

Major Comments

1. **The methodological contribution and the gap in the existing literature need to be stated much more precisely.**

   The manuscript repeatedly states that it "develops several Bayesian models." However, as currently presented, the three models appear to be nested specifications within one common hierarchical framework rather than substantially distinct Bayesian models. The baseline model combines a Poisson–Gamma hierarchy with standard Gaussian hierarchical effects, random-walk priors, an AR(2) temporal component, and interaction terms; the two expanded specifications then add higher-order interaction terms.

   I encourage the authors to clarify exactly what is new. It would be helpful to distinguish explicitly among: (i) a new statistical model or prior construction, (ii) a new computational strategy, and (iii) an application-specific integration of established Bayesian hierarchical components for highly disaggregated migration data. If the principal contribution is the construction of a sufficiently flexible and computationally feasible modeling framework for this application, that is a legitimate contribution, but it should be stated directly and the methodological claims should be calibrated accordingly.

   Relatedly, the introduction currently establishes mainly that the Bayesian literature on internal migration is limited. This is not yet a sufficiently specific methodological gap. The revised manuscript should compare the proposed framework directly with the closest existing approaches and explain which limitation of previous methods motivates each important modeling component. For example, are existing methods unable to accommodate single-year age profiles, origin–destination-specific heterogeneity, changing age profiles over time, very high-order interactions, or the required level of computational scalability? A concise comparison table

covering dimensions modeled, age smoothing, origin–destination structure, temporal dynamics, uncertainty quantification, and computational strategy would substantially strengthen the motivation.

2. The treatment of high dimensionality, regularization, and shrinkage requires a substantially deeper explanation.

The most complex specification contains 121,379 β parameters, in addition to 170,352 cell-specific rate quantities. Given that the dataset itself contains 170,352 observed cells, the effective regularization of the model is central to its statistical credibility.

Equation (3) gives a useful interpretation of shrinkage of the cell-specific rate γ_{ijast} toward its prior mean μ_{ijast}. However, this does not explain how the very large collection of β parameters, including the high-order interactions, is regularized. My reading is that the principal mechanism is structured smoothing and partial pooling through Gaussian priors, shared variance parameters, and RW1/RW2 specifications. This should be made explicit. In particular, the current model does not appear to perform sparse shrinkage or interaction selection in the usual sense; instead, it allows all specified interactions to enter while controlling their roughness and magnitude through hierarchical priors.

I suggest that the authors provide more evidence on the amount of regularization actually induced by the model. At minimum, posterior summaries of the major variance/smoothing parameters would be informative. Sensitivity to the half-normal hyperpriors would also be useful, especially for the high-order interaction terms. More generally, the authors should explain what prevents the most complex model from becoming a near-saturated smoother and should quantify, where possible, the effective complexity of the higher-order components. This issue is particularly important because the model-comparison procedure favors progressively richer interaction structures.

3. The computational inference procedure needs a clearer and more complete description.

Section 3.3 does not currently provide enough information to understand exactly how posterior inference is performed. The manuscript states that the implementation uses TMB, Laplace approximation, numerical integration, and analytic marginalization of the cell-specific rates, but it later refers to 1,000 posterior draws without explaining how these draws are obtained.

Please state clearly which quantities are optimized, which quantities are integrated using the Laplace approximation, and how posterior draws of the parameters, hyperparameters, and derived quantities are generated. If an MCMC step is used at any stage, appropriate convergence and sampling-efficiency diagnostics should be reported (for example, multiple-chain diagnostics, $\widehat{R}$, effective sample sizes, and, where relevant, ESS per second). If the reported posterior draws instead arise from a Gaussian/Laplace approximation, then standard MCMC diagnostics are not applicable; in that case, the authors should explain the approximation explicitly and provide some assessment of its accuracy, ideally through comparison with a more exact posterior computation on one or more reduced-size examples.

The reported runtimes are useful, but computational reporting could also be made more informative by specifying the exact processor, number of cores/threads, whether parallelization is used, peak memory consumption, and whether the reported times include generation of the posterior draws. Because computational scalability is emphasized as an advantage of the

approach, even a limited empirical assessment of scaling with the number of cells and/or model terms would be valuable.

Finally, the description of the analytic integration of γ_{ijast} as borrowing from "collapsed Gibbs sampling" may be confusing. As presented in the appendix, integrating the Gamma-distributed rates out of the Poisson–Gamma hierarchy yields a Negative-Binomial marginal likelihood directly. If no Gibbs sampler is actually used, the terminology should be revised so that conjugate marginalization is clearly distinguished from a collapsed Gibbs algorithm.

4. **Model comparison and, especially, forecasting validation need to be strengthened substantially.**

   Forecasting is central to the title, abstract, and conclusions, yet the manuscript currently does not provide an out-of-sample assessment of forecast performance. WAIC and posterior predictive checks evaluate aspects of fit to the observed 2012–2024 data, but they do not establish that the model forecasts future migration rates accurately. Consequently, the statement that the model "performs well in forecasting" is stronger than the empirical evidence currently provided.

   I strongly recommend a historical forecasting or rolling-origin validation exercise. For example, the authors could fit the model using data ending in earlier years and evaluate one- through five-year-ahead forecasts against subsequently observed data. Appropriate summaries could include RMSE or MAE, predictive log scores, empirical coverage of predictive intervals, interval width, and calibration across forecast horizons. The three proposed specifications should be compared in this exercise, together with at least one simple forecasting benchmark. Such an analysis would also provide a much more direct answer to whether the additional high-order interactions improve forecasting rather than merely in-sample fit.

   There is also an important issue regarding the current WAIC calculation. The manuscript appears to compute pointwise predictive densities conditional on posterior draws of the observation-specific latent rates γ_{ijast}, even though these rates are marginalized during fitting and are themselves updated using the corresponding observations. The authors should justify why this conditional WAIC is aligned with the intended predictive target. For prediction of new/future cells, a marginal criterion based on the Negative-Binomial likelihood $p(y_{ijast} \mid \mu_{ijast}, \xi)$ may be more relevant. In any case, time-structured out-of-sample validation would be preferable as the primary evidence for forecast performance.

   In addition, uncertainty in the model-comparison criterion should be reported. The manuscript currently reports only scalar WAIC values. Pointwise differences and a standard error (or an equivalent uncertainty summary) would help readers assess whether the difference between the first and second expanded models is practically meaningful.

Minor Comments

1. The geographic scope should be clarified. The title and manuscript repeatedly refer to the "UK," whereas the nine regions listed in Section 2 appear to be the nine standard regions of England. Please reconcile the title, abstract, data description, and interpretation with the actual geographic coverage of the analysis.

2. The parameterization of the Gamma distribution in Equation (2) should be stated explicitly. From the subsequent expressions, the second parameter appears to be a rate rather than a scale,

but this should not be left implicit.

3. The AR(2) specification requires clarification. Restricting each autoregressive coefficient individually to $(-1, 1)$ does not by itself guarantee stationarity of an AR(2) process. Please state whether the joint prior is restricted to the AR(2) stationarity region, particularly because the initialization and marginal-variance construction are described using Yule–Walker equations.

4. In Equation (17), the recursion is over time, but the range below the equation appears to be written as $a = 2, \ldots, A$. Presumably this should be $t = 2, \ldots, T$.

5. Please check the indexing of the origin–destination–age–time interaction around Equation (17). There appears to be an inconsistency in which a sex index is used in one location where a time index is expected.

6. Please check the notation for the exposure/population-at-risk quantity. It is initially defined as w_{iast}, since the same origin population is at risk of moving to multiple destinations, but variants with a destination index appear elsewhere.

7. The manuscript would benefit from a clearer discussion of identifiability and centering constraints for the large collection of main effects and interaction terms. Several components are anchored by fixing an initial element to zero, whereas others appear to rely on proper priors rather than explicit sum-to-zero constraints. The implications for interpretation of the individual effects should be explained.

8. The conditional independence assumption for destination-specific migration counts sharing the same origin–age–sex–year population at risk deserves some discussion. A multinomial or competing-destination representation would also be natural. If the independent Poisson formulation is intended as an approximation, a brief justification would be helpful.

9. The term "sparse" is used repeatedly. Since only a relatively small fraction of cells contain zero counts, it would be helpful to clarify that the main issue is low counts in many finely disaggregated cells rather than sparsity in the sense of a very large proportion of zeros.

10. The posterior predictive checks currently focus on the minimum, maximum, mean, and standard deviation of age-specific direct estimates. More migration-specific diagnostics could provide a more interpretable assessment of model adequacy, for example the location and magnitude of the student-migration peak, old-age behavior, temporal shocks, and the frequency of zero or very low counts.

11. The statement that a satisfactory model should have approximately 10% or fewer posterior predictive p-values outside the 0.05–0.95 interval is somewhat heuristic. Since posterior predictive p-values are generally not uniformly distributed, the interpretation of this threshold should be justified more carefully or softened.

12. The manuscript should quantify uncertainty in the forecast results more explicitly in the text rather than relying primarily on visual inspection of Figures 4–6.

13. The phrase "all three models have 170,352 rate parameters" may be slightly misleading because

these cell-specific rates are analytically marginalized during fitting and subsequently generated in post-processing. It would be clearer to distinguish latent rate quantities from parameters explicitly represented in the numerical optimization.

14. There is a typographical error in the keyword list: “hierarhical” should be “hierarchical.”

Overall, I believe the manuscript addresses a useful applied problem and has the potential to become a solid contribution, particularly if the authors sharpen the positioning of the methodological contribution and provide a convincing out-of-sample forecasting evaluation. The revisions above would substantially improve the statistical justification, transparency, and interpretation of the proposed framework.